

S.No. 615

NBS4101

No. of Printing Pages : 5

Following Paper ID and Roll.No. to be filled in your Answer Book.

Paper ID : 49901

Roll
No.

1	2	3	0	4	3	9	3	6	6
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B.Tech. Examination -2023-24
(Odd Semester)

MATRICES AND CALCULUS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION-A

1. Attempt each part in this section. Each part carry equal marks.

$$8 \times 1 = 8$$

- (a) Define orthogonal matrix.
- (b) Define rank of a matrix.
- (c) Find the n^{th} derivative of $\log x^2$.
- (d) State Euler's theorem on homogeneous function.

[P. T. O.

(e) Evaluate—

$$\int_0^a \int_0^x xy \, dy \, dx$$

(f) Evaluate—

$$\int_0^{\infty} \sqrt{x} e^{-x} \, dx$$

(g) Find the normal vector to the surface $z=2xy$ at $(2, 1, 4)$.(h) What is surface integral of a vector function f over the surface S .

SECTION-B

2. Attempt any two parts in this section. Each part carry equal marks. $2 \times 6 = 12$

(a) Verify Cayley Hamilton theorem for the

$$\text{matrix} \begin{bmatrix} 3 & 1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}. \text{ Hence find } A^{-1}.$$

(b) Find the value of the n th derivative of $y = e^{m \sin^{-1} x}$ for $x=0$.(c) Find the shortest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$.(d) Verify Stoke's theorem for the function $\vec{F} = x^2 \hat{i} - xy \hat{j}$ integrated round the square in the plane $z=0$ and bounded by the lines $x=0$, $y=0$, $x=a$, $y=a$.

SECTION-C

3. Attempt any two parts from each questions. Each part carry equal marks. $5 \times 8 = 40$

(a) Find the rank of the matrix—

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

by reducing it to normal form.

(b) For what value of λ the equations

$$x + y + z = 1$$

$$x + 2y + 4z = \lambda$$

$$x + 4y + 10z = \lambda^2$$

has a solution and solve them completely in each case.

- (c) Find the eigen values and corresponding eigen vectors of the matrix—

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

4. (a) Expand $e^x \sin y$ in powers of x and y upto terms of third degree.

- (b) If $y_1 = \frac{x_2 x_3}{x_1}$, $y_2 = \frac{x_3 x_1}{x_2}$, $y_3 = \frac{x_1 x_2}{x_3}$ then show that the Jacobian of y_1, y_2, y_3 with respect to x_1, x_2, x_3 is 4.

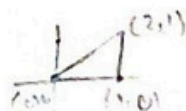
- (c) If $V = (x^2 + y^2 + z^2)^{m/2}$ then find the value of

$$m(m \neq 0) \text{ which will make } \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

5. (a) Evaluate $\int_0^2 \int_0^x dx dy$ by changing the order of integration.

- (b) Evaluate —

$$\int_0^{\log 2} \int_0^x \int_0^{x+y} e^{x+y+z} dx dy dz$$



- (c) Prove that—

$$\sqrt{n} \sqrt{n + \frac{1}{2}} 2^{2n-1} = \sqrt{\pi} \sqrt{2n}$$

where n is positive.

6. (a) Find the directional derivative of $\phi(x, y, z) = x^2 yz + 4xz^2$ at $(1, -2, 1)$ in the direction of $2\hat{i} - \hat{j} - 2\hat{k}$.

- (b) If $\vec{A} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$ evaluate the line integral $\oint \vec{A} \cdot d\vec{r}$ from $(0,0,0)$ to $(1,1,1)$ along the curve $c: x=t, y=t^2, z=t^3$.

- (c) Using Green's theorem evaluate $\oint (x^2 y dx + x^2 dy)$, where c is the boundary described counter clockwise of the triangle with vertices $(0,0), (1,0), (1,1)$.

